

SAFWAN ALI

AI Engineer • Data Scientist • LLM & Full-Stack AI Developer

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Karachi, Pakistan

PROFESSIONAL SUMMARY

AI Engineer and final-year Computer Science student (Salim Habib University, June 2026) with over a year of professional experience shipping production-grade AI systems at MarkyTech, where a three-month AI internship converted into an ongoing engineering role. Hands-on across the full AI development lifecycle — LLM-powered chatbots, RAG pipelines, real-time voice AI agents, agentic multi-agent workflows, computer vision and generative AI tooling, and containerised full-stack AI applications. Delivered a SUPARCO-aligned Final Year Project combining satellite imagery, deep learning, and LLMs for lunar soil analysis. Skilled at translating business requirements into scalable, deployable AI solutions using LangChain, AutoGen, FastAPI, NestJS, Next.js, and Docker-based deployment on GPU infrastructure.

PROFESSIONAL EXPERIENCE

MarkyTech

Aug 2025 – Present

Digital solutions agency specializing in social media marketing, web development, and AI-powered SaaS products. Progressed from AI Intern to AI Engineer within the same engagement.

AI Engineer (Part-time)

Oct 2025 – Present

Hybrid • Pakistan

- Design, build, and deploy end-to-end AI-driven SaaS features, bridging AI research and production tooling for the commercial market.
- Developed real-time **voice AI agents** with streaming speech-to-text and text-to-speech pipelines, handling low-latency audio streaming, interruption/barge-in handling, and multi-turn conversational state for automated client calls and voice-driven support.
- Engineered the lead generation component for Busman AI, an AI-powered business automation product, integrating conversational AI to qualify and capture leads through intelligent chat and voice flows.
- Built and deployed LLM-powered chatbots for clients using OpenAI GPT, Gemini, and Claude APIs, applying LangChain for orchestration, context management, and retrieval-augmented generation (RAG) over vector-indexed knowledge bases.
- Implemented agentic multi-agent workflows with AutoGen and AgentScope to automate complex, multi-step business processes, and fine-tuned and optimised open-source models for latency, cost, and reliability.
- Developed computer vision and generative AI tooling, including hyper-realistic image and video generation engines served on GPU infrastructure (RunPod) for industry-specific use cases.
- Built production REST APIs with FastAPI and NestJS serving model inference endpoints, backed by PostgreSQL and MySQL, with Next.js frontends delivering end-to-end AI web applications.
- Containerised AI services with Docker and Podman and automated build and release workflows with Git and GitHub Actions, improving deployment repeatability across environments.
- Implemented Crawl4AI scraping and data ingestion pipelines feeding structured business intelligence into AI systems, and automated internal Python tasks for data processing, reporting, and workflow orchestration.

AI Intern

Aug 2025 – Oct 2025

Karachi, Sindh, Pakistan

- Completed a rigorous three-month Artificial Intelligence program, collaborating with MarkyTech's technical team to implement AI-driven solutions across internal projects and meeting every technical milestone on schedule.
- Built working knowledge of production AI architectures and real-world deployment practices, leading to conversion into an engineering role.

EDUCATION

Bachelor of Science in Computer Science

2022 – June 2026

Salim Habib University — Karachi, Pakistan — Specialization: Data Science, Machine Learning & AI

- Final Year Project: AI-Driven Lunar Soil Composition Analysis System — developed in alignment with SUPARCO's ICUBE-Qamar lunar satellite mission.
- Relevant coursework: Machine Learning, Deep Learning, Database Systems, Statistics & Probability, Data Structures & Algorithms.

KEY PROJECTS

AI-Driven Lunar Soil Composition Analysis (FYP — SUPARCO)

Python · SAM 2.1 · ResNet-18 · Gemini 2.0 Flash · Streamlit

- Built an end-to-end pipeline analyzing 457 real Chang'e 3 Yutu rover images to estimate 6 lunar mineral oxides (FeO, TiO₂, MgO, SiO₂, Al₂O₃, CaO), calibrated to Chang'e 3 APXS ground-truth data and aligned with Pakistan's ICUBE-Qamar satellite sensor specifications.
- Integrated SAM 2.1 for terrain segmentation and a ResNet-18 deep learning classifier to identify geological units; automated scientific report generation via the Gemini 2.0 Flash LLM.
- Delivered a Streamlit Mission Control dashboard with real-time telemetry and physics-based geochemical constraints (oxide-sum conservation).

[GitHub](#)

Heart Disease Prediction — Random Forest Classifier

Python · Scikit-learn · Streamlit (Deployed)

- Developed and deployed a Random Forest classifier for clinical heart disease risk prediction with feature selection, data preprocessing, and hyperparameter tuning.
- Shipped as a live Streamlit Cloud application serving real-time inference.

[GitHub](#) | [Live Demo](#)

Fruit Classification CNN

Python · TensorFlow · Keras

- Built a CNN with data augmentation and batch normalization for multi-class fruit image classification; applied transfer learning and hyperparameter tuning to maximize validation accuracy.

[GitHub](#)

Customer Segmentation — K-Means Clustering

Python · Scikit-learn · Matplotlib

- Implemented K-Means clustering on customer purchasing data with elbow-method optimization, producing actionable marketing segment visualizations.

[GitHub](#)

Additional ML & Data Science Work

Python · Scikit-learn · NLP · Jupyter

- SVM Handwritten Digit Classifier (MNIST) with HOG feature extraction; Linear Regression sales-forecasting model quantifying advertising spend impact across TV, radio, and print channels; plus NLP spam detection and GAN experiments across a 30-repository public portfolio.

[GitHub](#)

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

LLM & Generative AI: LangChain, AutoGen, AgentScope, OpenAI API, Gemini API, Claude API, Hugging Face, RAG Pipelines, Vector Search & Embeddings, Prompt Engineering, Agentic Workflows, Voice AI Agents (STT/TTS), Open-Source LLM Fine-Tuning

ML / Deep Learning: TensorFlow, Keras, Scikit-learn, XGBoost, CNNs, SVMs, Random Forest, K-Means, SAM 2.1, ResNet, Computer Vision, NLP, Model Evaluation & Hyperparameter Tuning

Data Engineering & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Crawl4AI, ETL & Data Ingestion Pipelines, Jupyter Notebook, Anaconda

Backend & APIs: FastAPI, NestJS, Node.js, Express.js, REST APIs, WebSockets, Microservices

Frontend: Next.js, React, Tailwind CSS, Streamlit

Databases: PostgreSQL, MySQL, MongoDB

DevOps, Cloud & MLOps: Docker, Podman, RunPod (GPU Compute), GitHub Actions, CI/CD, Git, GitHub, Linux, Nginx, Vercel, Model Deployment & Monitoring

Languages (Human): English (Professional), Urdu (Native)

ACHIEVEMENTS & NOTABLE WORK

- Converted from a three-month AI internship into an ongoing AI Engineer role at MarkyTech, demonstrating rapid professional growth in a production environment.
- Delivered a core lead generation component and production voice AI agents for Busman AI, a commercial AI-powered SaaS product, while still a final-year student.
- SUPARCO FYP alignment — Final Year Project designed for direct application to Pakistan's ICUBE-Qamar lunar satellite mission, one of Pakistan's most significant space programs.
- 30 public GitHub repositories spanning deep learning, computer vision, NLP, agentic AI, clustering, and full-stack AI applications — github.com/Safwan2003.